Implementation Progress Report

How each step works — an illustrated walkthrough

Predicting the Zero-Shot Transfer of a Promptable Video Tracker (SAM 3 on SA-FARI)

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This report explains, in plain terms and with pictures, *how* each part of the system works and what is built so far. The one-line goal: given a new animal in a new place, **predict how well SAM 3 will track it *before* we run it**, and explain *why* — separately for **finding** the animal (`pDetA`) and **following** it (`pAssA`).

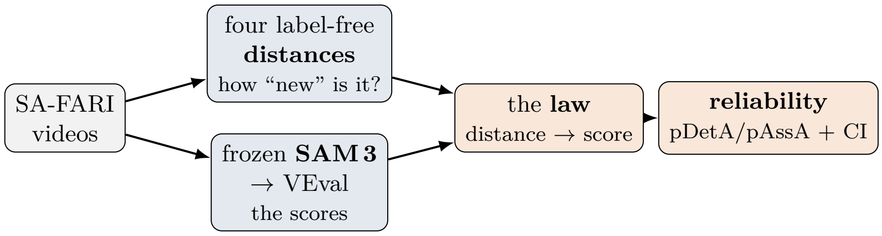

Figure 1: The whole system. The **distances** (top) are the inputs X; the frozen **SAM 3** scores (bottom) are the outputs Y; we fit a small **law** $X \to Y$ and turn it into a **reliability** estimate. The distance side is complete; the score side is built and awaiting a GPU run.

Where we are	
done	data acquisition; dataset/schema; the two experiments + frozen reference; all four label-free distances (taxonomic, temporal, visual, environment); the shared embedding pipeline.
built	SAM 3 inference harness + VEval scoring (run on Colab; weights access confirmed).
next	SAM 3 familiarity proxy; feature-table assembly; fit the law, cross-validation, reliability tool.

1 Step 1 — Getting the data

Everything is built on **SA-FARI**: thousands of short camera-trap videos of wild animals, each frame hand-outlined. Four lists describe it, linked by ids (Fig. 2):

- **videos** — one entry per clip, with its frame files, its camera *location*, and *when* it was filmed.
- **categories** — a big dictionary of text concepts ("a monkey", "ocelot", …), each with a `category_id`. The real animals carry a full *tree-of-life* classification; most entries are generic distractors with no classification.
- **annotations** — the ground-truth outlines (masks), one per frame, for the animals that are actually present.
- **probes** (`video_np_pairs`) — the questions we ask: "is *this animal* in *this video*?". Many are **hard negatives** — the animal is *absent*, and a good model should return nothing.

We fetch the outlines from Hugging Face (a gated download) and pull only the frames we need from a public Google bucket, on demand. A key early correction: the animal's identity is the `category_id`, not the raw prompt text (two different prompts can mean the same species), so everything keys on `category_id`.

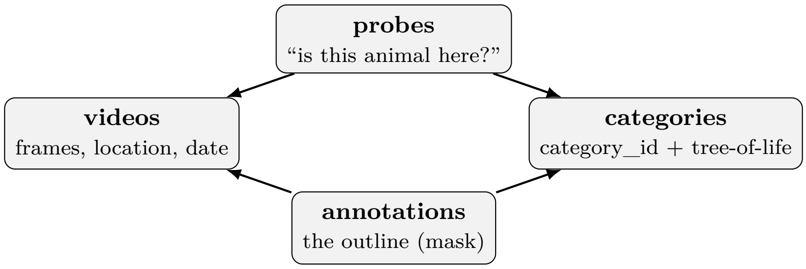

Figure 2: How SA-FARI is wired together. A *probe* asks whether a named animal (a `category_id`) appears in a video; *annotations* give the ground-truth outlines for the ones that do.

2 Step 2 — The two experiments

The whole idea rests on a *reference* (“what we’ve seen”) and a *probe* (“the new thing”): a distance measures how far a probe is from the reference. Inspecting the data changed our plan here. The official split holds out *new places* but re-uses the *same species* on both sides — so on that split “species novelty” is always zero. Because SAM 3 is **frozen** (it never trains on any of this), the split is only *our* choice of reference, so we are free to re-slice it. We therefore run **two complementary experiments** (Fig. 3):

- **Species hold-out** (primary). Hide one species at a time and ask “how novel is it versus *all the others*?” — *leave-one-species-out*. This is the detection/novelty test.
- **Location hold-out** (the official split). Reference = one region, probe = a totally different region. This is the environment/association test.

The *leakage firewall* is simply a closed-book rule: when we score how novel species X is, X is never allowed into its own reference — otherwise it would trivially look “distance 0 from itself”. A test checks this.

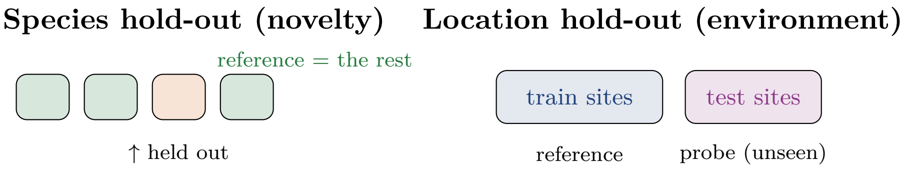

Figure 3: Two ways to slice the same data. **Left**: each species is held out in turn and compared to the others (species varies, place stays familiar). **Right**: the official geographic split (place varies, species stays familiar). Together they isolate the two hypotheses.

3 Step 3 — The four “how new is it?” distances

Each distance answers a plain question and is computed with *no label of the target animal.*

Taxonomic — how far apart on the tree of life?

Count the steps up the family tree until two animals share an ancestor: same genus = 1, same family = 2, same order = 3, and so on (Fig. 4). For each species we take the distance to its *nearest other* species. Across the 72 fully-classified species this spreads from 0 to 4 (Fig. 5) — most animals have a same-family neighbour, but a fifth are clearly novel (≥ 3). The two 0s are a quirk: “pig” and “wild boar” are the *same* species under two names. This spread is what makes the novelty axis testable.

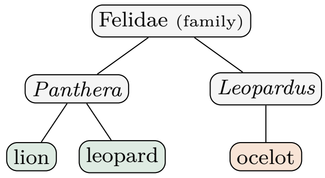

Figure 4: Lion ↔ leopard share a genus (distance 1); lion ↔ ocelot share only the family (distance 2).

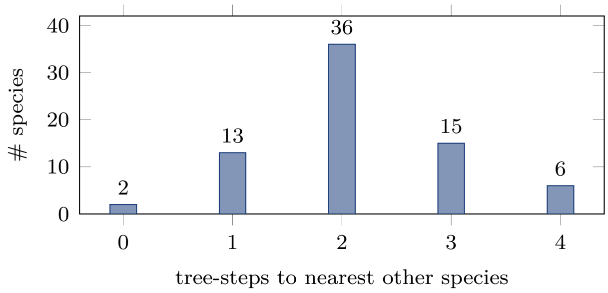

Figure 5: Measured leave-one-species-out distances (72 species). Real numbers from our run.

Temporal — how many years apart?

The simplest one: the number of years between the probe's footage and the nearest reference footage (2014 — 2024). A 2023 clip against a 2015 reference is a gap of 8.

Visual — how unfamiliar does the animal *look*?

We cut the animal out of a few frames using its outline (background blacked out, so we measure the *animal*, not the bush behind it), pass each cut-out through a frozen vision model (DINOv2) to get a numeric "fingerprint", and average a species' fingerprints into one **prototype**. The distance is how different a species' prototype is from the nearest *other* prototype (cosine distance; Fig. 6). On a six-species check the spread is sensible — the pig looks most distinct, the smaller mammals cluster together (Fig. 7).

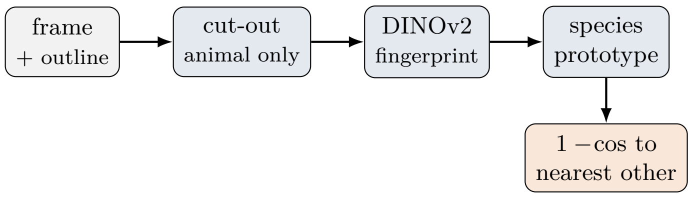

Figure 6: The visual distance, end to end. "Environment" is the same picture with the animal *erased* instead of cut out, grouped by location instead of species.

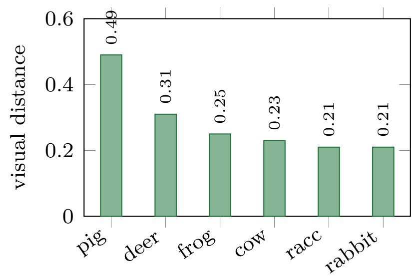

Figure 7: Measured visual distances (six-species subset). Real numbers from our run.

Environment — how unfamiliar does the *place* look?

Same trick for the scene: *erase* the animal (fill its pixels with the average background) so only the habitat remains, fingerprint that, and average per camera site into a "typical scene". The distance is how far a probe site looks from the nearest reference site. A bonus flag spots **night / infrared** footage by noticing the frame is basically grey. On the location hold-out the distances ran roughly 0.06–0.44, and the infrared night sites were flagged correctly while the daytime colour sites were not.

4 Step 4 — Measuring SAM 3 (the scores we predict)

This is the "truth" we compare against. **SAM 3** is a *promptable* tracker: give it a text prompt ("ocelot") and it finds and follows every matching animal across the video — with no training on that species. The official evaluator splits its score into the two things we care about (Fig. 8):

- `pDetA` — **detection**: did it *find* the animals (and not hallucinate on the hard negatives)?
- `pAssA` — **association**: did it keep *each identity* consistent over time?

Our harness runs this over every probe and writes one small file per video, so a run can stop and resume without losing work (important on Colab, where sessions time out). The evaluator then turns the outlines into the two numbers per cell. *This code is built and tested with a stand-in; the real run happens on a GPU.*

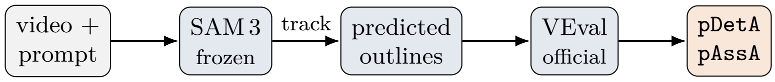

Figure 8: Measuring the ground truth. A prompt goes into frozen SAM 3, which outlines and tracks matches; the official scorer turns that into *detection* and *association* numbers per cell.

5 Step 5 — Fitting the law (planned)

With the distances (X) and scores (Y) in hand, the last stage fits a small, honest model:

- **A curve, not a black box.** A beta regression (a straight line through a "squashing" transform, because scores live in $[0, 1]$) maps distances to a predicted score — one model for detection, one for association.
- **Which factor matters?** Variance partitioning asks whether *detection* is driven by *species novelty* and *association* by *environment* — the headline contrast.
- **Does it generalise?** We hide whole species (and whole places), predict them from distance *alone*, and check the prediction lands near the truth, with honest error bars (Fig. 9). Success is a *validated* predictor, not a higher score.
- **The tool.** Four distances in → predicted `pDetA`/`pAssA` + a confidence interval out: the practical "will it work here?" answer.

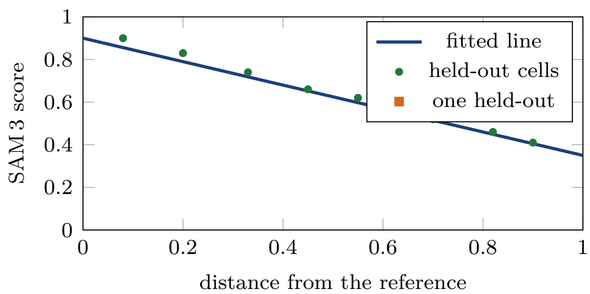

Figure 9: The target picture (*illustrative — pending the real scores*): predict a held-out cell's score from its distance and check it lands on the line, with bootstrap confidence intervals.

6 Where we are, and the next step

The **distance side is finished and sanity-checked** on real data, and the **measurement side is built**. The codebase is typed, configuration-driven, lint-clean, and covered by a growing test suite; the label-free features were refactored and verified to reproduce their outputs exactly. The immediate milestone is the **first decision gate**: run frozen SAM 3 over the test split on Colab (weights access is confirmed), score it, and check the numbers sit in SA-FARI's published SAM 3 range. Passing that gate opens the modelling half — fitting the detection-versus-association law and building the reliability estimator.